Type MS · Velocity and displacement by integrating

Answer 1 of 8 · Question 5 part c · 4 marks

2022 JAN · Paper 02 · extract 1 of 3 · find part c in this mark scheme extract

5(a)	$3t^2-23t+30=0\Rightarrow(3t-5)(t-6)=0\Rightarrow t=6,\quad \frac{5}{3}$	M1A1A1 [3]
(b)	$a=\frac{\mathrm{d}v}{\mathrm{d}t}=6t-23$ $(6t-23>0)\Rightarrow t>\frac{23}{6}$	M1 A1 [2]
(c)	$s=\int 3t^2-23t+30\ \mathrm{d}t=\frac{3t^3}{3}-\frac{23t^2}{2}+30t+c$ $26=\frac{3(8)^3}{3}-\frac{23(8)^2}{2}+30(8)+c\Rightarrow c='10'$ $d=0+0+0+'10'\Rightarrow d=10$	M1A1 M1 A1 [4]
	Total	**9 marks**

Type MS · Velocity and displacement by integrating

Answer 1 of 8 · Question 5 part c · 4 marks

2022 JAN · Paper 02 · extract 2 of 3 · find part c in this mark scheme extract

Part	Mark	Notes
5 (a)	M1	For setting the given expression for $v = 0$ and attempting to solve the 3TQ See general guidance for the definition of an attempt $3t^2 - 23t + 30 = 0 \Rightarrow (3t-5)(t-6) = 0 \Rightarrow t = ..., ...$
	A1	For either $t = 6$ or $t = \frac{5}{3}$
	A1	For both $t = 6$ and $t = \frac{5}{3}$
(b)	M1	For differentiating the given expression for v which must be correct for this mark. $a = \frac{dv}{dt} = 6t - 23$
	A1	$(6t - 23 > 0) \Rightarrow t > \frac{23}{6}$ Accept equivalent exact values including $t > 3.8\dot{3}$
(c)	M1	For attempting to integrate the given expression for v See General Guidance for the definition of an attempt with no terms differentiated. $+c$ is not required for the award of this mark. $s = \int 3t^2 - 23t + 30 \, dt = \frac{3t^3}{3} - \frac{23t^2}{2} + 30t + c$
	A1	For the correct integrated expression, which must include a constant term e.g.+ c $s = \frac{3t^3}{3} - \frac{23t^2}{2} + 30t + c$ Simplification is not required for this mark
	M1	For substituting the given values of $t = 8$ when $s = 26$ into their **integrated** expression to find c $26 = \frac{3(8)^3}{3} - \frac{23(8)^2}{2} + 30(8) + c \Rightarrow c = '10'$ $[d = 0 + 0 + 0 + '10' \Rightarrow d = ...]$
	A1	For the correct value of $d = 10$

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Answer 1 of 8 · Question 5 part c · 4 marks

2022 JAN · Paper 02 · extract 3 of 3 · find part c in this mark scheme extract

Question	Scheme	Mark

Type MS · Velocity and displacement by integrating

Answer 2 of 8 · Question 3 part a.ii · 4 marks

2023 NOV · Paper 02 · extract 1 of 3 · find part a.ii in this mark scheme extract

Question	Scheme	Marks
3(a)(i)	$v = 6t^2 - 16t + c$	M1
	$t = 0, \quad v = 12$	A1
	$v = \frac{6t^2}{2} - 16t + 12 = \left[3t^2 - 16t + 12\right]$	M1
(ii)	$s = \frac{3t^3}{3} - \frac{16t^2}{2} + 12t + k$	A1
	$\left[t = 0, \quad s = 0\right] \Rightarrow k = 0$	[4]
	$s = t^3 - 8t^2 + 12t$	
(b)	At the origin, $s = 0$	
	$t^3 - 4t^2 + 12t = 0$	M1
	$\Rightarrow t(t-2)(t-6) = 0$	dM1
	$\Rightarrow t = 2, 6, (0)$	A1
	P first returns to the origin when $t = 2$ seconds	[3]
		Total 7 marks

Type MS · Velocity and displacement by integrating

Answer 2 of 8 · Question 3 part a.ii · 4 marks

2023 NOV · Paper 02 · extract 2 of 3 · find part a.ii in this mark scheme extract

Part	Mark	Notes
(a)(i)	M1	For integrating the given expression. ($+c$) not required for this mark See General Guidance for the definition of an attempt. No term to be differentiated.
	A1	Finds the value of c **and** the correct expression for v
(ii)	M1	For an acceptable attempt to integrate **their** v. See General Guidance for the definition of an attempt. ($+k$) not required for this mark. No term to be differentiated.
	A1	For the correct expression for the distance, which must have come from establishing **explicitly** that $k = 0$ from the integration. Without $+ k$ in their integration this mark is automatically lost.
(b)	M1	For setting their expression [which must be a cubic] for $s = 0$
	dM1	For solving the cubic Minimally acceptable attempt is $\Rightarrow t(t \pm A)(t \pm B) = 0$ where $\lvert AB \rvert = 12$ or also allow $(t \pm A)(t \pm B) = 0$ where $\lvert AB \rvert = 12$ as candidates are told that one value of $t = 0$ Any method to solve the quadratic is acceptable. Formula or completing the square. This is a dependent mark. <u>Use of a calculator</u> If there is no working seen and either their values of t are incorrect or their s is incorrect, only award this mark if a correct method is used/seen to solve the equation.
	A1	For the correct value of $t = 2$ explicitly stated.

Type MS · Velocity and displacement by integrating

Answer 2 of 8 · Question 3 part a.ii · 4 marks

2023 NOV · Paper 02 · extract 3 of 3 · find part a.ii in this mark scheme extract

Question	Scheme	Marks

Type MS · Velocity and displacement by integrating

Answer 3 of 8 · Question 4 part a · 3 marks

2019 JUN · Paper 02 (regional R) · extract 1 of 2 · find part a in this mark scheme extract

Number		
4 (a)	$v = \int (6t-12)\ \mathrm{d}t = 3t^2 - 12t\ (+c)$ When $t = 0$, $v = 0$ so $c = 0$ $v = 3t^2 - 12t = 3\times 2^2 - 12\times 2 = -12\ (\text{m/s})$	M1A1 A1cso [3]
(b)	$v = 0,\ 3t^2 - 12t = 0 \Rightarrow 3t(t-4) = 0 \Rightarrow T = 4$	M1A1 [2]
(c)	$s = \int (3t^2 - 12t)\ \mathrm{d}t = t^3 - 6t^2\ (+k)$ $d = \left\lvert \left[t^3 - 6t^2\right]_0^4 \right\rvert + \left[t^3 - 6t^2\right]_4^8 = 32 + 160 = 192\ (\text{m})$	M1 M1A1 [3]
Alternative		
(c)	$s = \int (3t^2 - 12t)\ \mathrm{d}t = t^3 - 6t^2\ (+k)$ At time $t = 0$, distance = 0, $k = 0$ $t = 4 \quad s = 4^3 - 6\times 4^2 = 64 - 96 = -32$ $t = 8 \quad s = 8^3 - 6\times 8^2 = 512 - 384 = 128$ Total distance travelled = 32 +32 +128 = 192 (m)	M1 M1 A1
		Total 8 marks

(a)	M1	Attempt to integrate $6t - 12$, must see power of t increased by one in either term and no decrease in powers. Do not accept $v = t(6t-12)$.
	A1	Correct result of integration. Constant is not required.
	A1cso	Correct velocity. Negative sign is required. Must show or state constant = 0.
(b)	M1	$3t^2 - 12t = 0$ and attempt to solve. ft their integration if M1 awarded in (a).
	A1	$T = 4$ Allow $t = 4$. Do not allow if $T = 0$ is also given as an option.
(c)	M1	Attempt to integrate their velocity providing it includes at least one term in t^2 or higher powers. Must see power of t increased by one in at least one term and no decrease in powers. Do not accept $s = t\times$ their v.
	M1	Attempt to use limits of 0 to 4 and 4 to 8 and then add the magnitudes of the two values. ft their $T = 4$ if M1 awarded in (b).
	A1	192
Alternative		
(c)	M1	Attempt to integrate their velocity providing it includes at least one term in t^2 or higher powers. Must see power of t increased by one in at least one term and no decrease in powers. Do not accept $v = t(6t-12)$.
	M1	Substitute and attempt to find distance when $t = 4$ and when $t = 8$. Working to confirm $k = 0$ must be shown. ft their $T = 4$ if M1 awarded in (b).
	A1	192

Type MS · Velocity and displacement by integrating

Answer 3 of 8 · Question 4 part a · 3 marks

2019 JUN · Paper 02 (regional R) · extract 2 of 2 · find part a in this mark scheme extract

Question number	Scheme	Marks

Type MS · Velocity and displacement by integrating

Answer 4 of 8 · Question 5 part a · 3 marks

2024 JUN · Paper 01 · extract 1 of 2 · find part a in this mark scheme extract

5(a)	$\left(v=\int(3t-4)\,\mathrm{d}t=\right)\frac{3t^2}{2}-4t+c$ (When $t=0$, $v=0 \Rightarrow c=0$) $\left(v=\right)\frac{3\times4^2}{2}-4\times4=8$ [m/s]	M1 M1A1 [3]
(b)	$0=\frac{3t^2}{2}-4t \Rightarrow t\left(\frac{3t}{2}-4\right)=0 \Rightarrow \frac{3t}{2}=4 \Rightarrow t=\frac{8}{3}$	M1A1 [2]
(c)	$\left(x=\int\left(\frac{3t^2}{2}-4t\right)\mathrm{d}t=\right)\frac{3t^3}{2\times3}-\frac{4t^2}{2}(+c)=\left[\frac{t^3}{2}-2t^2+c\right]$ When $t=0$ P is at the point with coordinates $(-10, 0)$ $\Rightarrow c=-10$ $\Rightarrow \left(x=\right)\frac{3^3}{2}-2\times3^2-10=-\frac{29}{2}$ (m)	M1 M1 dM1A1 [4]
		Total 9 marks

Part	Mark	Notes
(a)	M1	For a minimally acceptable attempt (see general guidance) to integrate the given expression. **At least one term correctly integrated.** Terms do not need to be simplified to attain this mark. No power of t to decrease.
	M1	For substituting the value of $t = 4$ into their changed expression (general rule of marking unless precluded, this method mark may be implied from a correct answer, unless from incorrect working).
	A1	For $(v =)8$
We do not need to see the calculation of $c = 0$ Note: substitution of $t = 4$ into the expression for a also gives an answer of 8 – please watch out for this and do not mark this correct.		
(b)	M1	For setting their changed expression = 0 and a **fully correct** attempt to solve, leading to a value for t. Note a correct vale of t can imply this mark. If the quadratic is not the correct quadratic, method must be shown.
	A1	For the value of $t=\frac{8}{3}$, ignore $t = 0$ is given. Accept answers which round to 2.7 or clear indication of 2.6 recurring.
(c)	M1	For a **minimally acceptable attempt** (see general guidance) to integrate their expression for v, which must be a **minimum 2 term** expression. Terms don't need to be simplified at this point. c does not need to be present. No power of t to decrease.
	M1	For correct substitution, into their changed expression, of $t = 0$ and $P = -10$ to find the value of c. -10, if seen at any point will usually imply this mark. The mark is for substitution of the correct values, to find c, if they rearrange incorrectly, the mark can still be awarded.
	dM1	For substituting the value of $t = 3$ into their changed expression, with their c. Dependent on the previous method mark.
	A1	For the displacement of $-\frac{29}{2}$ If distance of $\frac{29}{2}$ is given A0, no isw here.

Type MS · Velocity and displacement by integrating

Answer 4 of 8 · Question 5 part a · 3 marks

2024 JUN · Paper 01 · extract 2 of 2 · find part a in this mark scheme extract

Question	Scheme	Marks

Type MS · Velocity and displacement by integrating

Answer 5 of 8 · Question 5 part c · 4 marks

2024 JUN · Paper 01 · extract 1 of 2 · find part c in this mark scheme extract

5(a)	$\left(v=\int(3t-4)\,\mathrm{d}t=\right)\frac{3t^2}{2}-4t+c$ (When $t=0$, $v=0 \Rightarrow c=0$) $\left(v=\right)\frac{3\times4^2}{2}-4\times4=8$ [m/s]	M1 M1A1 [3]
(b)	$0=\frac{3t^2}{2}-4t \Rightarrow t\left(\frac{3t}{2}-4\right)=0 \Rightarrow \frac{3t}{2}=4 \Rightarrow t=\frac{8}{3}$	M1A1 [2]
(c)	$\left(x=\int\left(\frac{3t^2}{2}-4t\right)\mathrm{d}t=\right)\frac{3t^3}{2\times3}-\frac{4t^2}{2}(+c)=\left[\frac{t^3}{2}-2t^2+c\right]$ When $t=0$ P is at the point with coordinates $(-10, 0)$ $\Rightarrow c=-10$ $\Rightarrow \left(x=\right)\frac{3^3}{2}-2\times3^2-10=-\frac{29}{2}$ (m)	M1 M1 dM1A1 [4]
		Total 9 marks

Part	Mark	Notes
(a)	M1	For a minimally acceptable attempt (see general guidance) to integrate the given expression. **At least one term correctly integrated.** Terms do not need to be simplified to attain this mark. No power of t to decrease.
	M1	For substituting the value of $t = 4$ into their changed expression (general rule of marking unless precluded, this method mark may be implied from a correct answer, unless from incorrect working).
	A1	For $(v =)8$
We do not need to see the calculation of $c = 0$ Note: substitution of $t = 4$ into the expression for a also gives an answer of 8 – please watch out for this and do not mark this correct.		
(b)	M1	For setting their changed expression = 0 and a **fully correct** attempt to solve, leading to a value for t. Note a correct vale of t can imply this mark. If the quadratic is not the correct quadratic, method must be shown.
	A1	For the value of $t=\frac{8}{3}$, ignore $t = 0$ is given. Accept answers which round to 2.7 or clear indication of 2.6 recurring.
(c)	M1	For a **minimally acceptable attempt** (see general guidance) to integrate their expression for v, which must be a **minimum 2 term** expression. Terms don't need to be simplified at this point. c does not need to be present. No power of t to decrease.
	M1	For correct substitution, into their changed expression, of $t = 0$ and $P = -10$ to find the value of c. -10, if seen at any point will usually imply this mark. The mark is for substitution of the correct values, to find c, if they rearrange incorrectly, the mark can still be awarded.
	dM1	For substituting the value of $t = 3$ into their changed expression, with their c. Dependent on the previous method mark.
	A1	For the displacement of $-\frac{29}{2}$ If distance of $\frac{29}{2}$ is given A0, no isw here.

Type MS · Velocity and displacement by integrating

Answer 5 of 8 · Question 5 part c · 4 marks

2024 JUN · Paper 01 · extract 2 of 2 · find part c in this mark scheme extract

Question	Scheme	Marks

Type MS · Velocity and displacement by integrating

Answer 6 of 8 · Question 8 part whole · 8 marks

2020 JAN · Paper 01 (regional R) · extract 1 of 2 · find part whole in this mark scheme extract

8	$s = \int \left(3 + 5t - 2t^2\right) \mathrm{d}t = 3t + \frac{5t^2}{2} - \frac{2t^3}{3} + c$	M1
	when $t = 0$ $s = 5$	
	$5 = 0 + 0 - 0 + c$	B1
	$s = 5 + 3t + \frac{5t^2}{2} - \frac{2t^3}{3}$	A1
	When $s = x$	
	$\frac{\mathrm{d}x}{\mathrm{d}t} = 3 + 5t - 2t^2 = 0 \Rightarrow (2t+1)(t-3) = 0 \Rightarrow t = 3$	M1A1
	$\Rightarrow x = 5 + 3 \times 3 + \frac{5 \times 3^2}{2} - \frac{2 \times 3^3}{3} = \frac{37}{2}$ oe	A1
	$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = 5 - 4t$ when $t = 3$ $\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -7 \Rightarrow$ max	M1A1
		[8]
		Total 8 marks

Type MS · Velocity and displacement by integrating

Answer 6 of 8 · Question 8 part whole · 8 marks

2020 JAN · Paper 01 (regional R) · extract 2 of 2 · find part whole in this mark scheme extract

Question number	Scheme	Marks

Type MS · Velocity and displacement by integrating

Answer 7 of 8 · Question 2 part b · 5 marks

2022 JUN · Paper 01 (regional R) · extract 1 of 5 · find part b in this mark scheme extract

Number		
2(a)	$(a=)2t-3$	M1 A1 [2]
(b)	$(s=\int(t^2-3t+4)\mathrm{d}t)$	
	$(s=)\frac{t^3}{3}-\frac{3}{2}t^2+4t(+C)$ oe	M1 A1
	$(t=2, s=7)$	
	$\frac{2^3}{3}-\frac{3}{2}\times2^2+4\times2+C=7\Rightarrow C=\frac{7}{3}$	dM1
	$\frac{4^3}{3}-\frac{3}{2}\times4^2+4\times4+'\frac{7}{3}'=\frac{47}{3}$ oe	M1A1
		[5]
	ALT	
	Displacement $-7=\int_2^4\left(x^2-3x+4\right)\mathrm{d}x$	
	$=\left[\frac{x^3}{3}-\frac{3x^2}{2}+4t\right]_2^4$	[M1A1
	$=\left[\frac{4^3}{3}-\frac{3\times4^2}{2}+4\times4\right]-\left[\frac{2^3}{3}-\frac{3\times2^2}{2}+4\times2\right]$	dM1M1
	$=\frac{47}{3}$	A1]
		Total 7 marks

Type MS · Velocity and displacement by integrating

Answer 7 of 8 · Question 2 part b · 5 marks

2022 JUN · Paper 01 (regional R) · extract 2 of 5 · find part b in this mark scheme extract

Part	Mark	Additional Guidance
2(a)	M1	For a minimally acceptable attempt at differentiation (see general guidance), no power of t must increase.
	A1	Correct expression with or without $a = \ldots$
(b)	M1	For a minimally acceptable attempt at integration (see general guidance), no power of t must decrease. + C is not necessary for this mark.
	A1	Fully correct integration. C is not necessary for this mark.
	dM1	Correct substitution of $t = 2$ into **their** integrated expression, correctly equated to 7 and an attempt to rearrange to find C. **Note: This mark is dependent on the first M mark in (b)**
	M1	Substitution of $t = 4$ correctly into their expression for s, provided it is a changed expression from v
	A1	Correct value oe must be an exact value.
	ALT	
	M1	For a minimally acceptable attempt at integration (see general guidance), no power of t must decrease.
	A1	Fully correct integration
	dM1	For the correct limits between $t = 4$ and $t = 2$ **and** equates to $d - 7$ or equivalent
	M1	Substitutes the limits to evaluate the integral
	A1	Correct value oe must be an exact value.

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Answer 7 of 8 · Question 2 part b · 5 marks

2022 JUN · Paper 01 (regional R) · extract 3 of 5 · find part b in this mark scheme extract

Question number	Scheme	Marks

Type MS · Velocity and displacement by integrating

Answer 7 of 8 · Question 2 part b · 5 marks

2022 JUN · Paper 01 (regional R) · extract 4 of 5 · find part b in this mark scheme extract

Part	Mark	Additional Guidance
2(a)	M1	For a minimally acceptable attempt at differentiation (see general guidance), no power of t must increase.
	A1	Correct expression with or without $a = \ldots$
(b)	M1	For a minimally acceptable attempt at integration (see general guidance), no power of t must decrease. + C is not necessary for this mark.
	A1	Fully correct integration. C is not necessary for this mark.
	dM1	Correct substitution of $t = 2$ into **their** integrated expression, correctly equated to 7 and an attempt to rearrange to find C. **Note: This mark is dependent on the first M mark in (b)**
	M1	Substitution of $t = 4$ correctly into their expression for s, provided it is a changed expression from v
	A1	Correct value oe must be an exact value.
	ALT	
	M1	For a minimally acceptable attempt at integration (see general guidance), no power of t must decrease.
	A1	Fully correct integration
	dM1	For the correct limits between $t = 4$ and $t = 2$ **and** equates to $d - 7$ or equivalent
	M1	Substitutes the limits to evaluate the integral
	A1	Correct value oe must be an exact value.

Type MS · Velocity and displacement by integrating

Answer 7 of 8 · Question 2 part b · 5 marks

2022 JUN · Paper 01 (regional R) · extract 5 of 5 · find part b in this mark scheme extract

Question number	Scheme	Marks

Type MS · Velocity and displacement by integrating

Answer 8 of 8 · Question 7 part b · 5 marks

2023 JAN · Paper 01 (regional R) · extract 1 of 3 · find part b in this mark scheme extract

7(a)	$v = 1^2 - 10\times1 + 28 = 19$ [m/s]	B1 [1]
(b)	$s = \int\left(t^2 - 10t + 28\right)\,\mathrm{d}t = \frac{t^3}{3} - \frac{10t^2}{2} + 28t\,(+c)$ $24 = \frac{3^3}{3} - \frac{10\times3^2}{2} + 28\times3 + c \Rightarrow c = -24 \Rightarrow \left[s = \frac{t^3}{3} - \frac{10t^2}{2} + 28t - 24\right]$ $t = 5, \quad s = \frac{5^3}{3} - \frac{10\times5^2}{2} + 28\times5 - 24 = \frac{98}{3}$ [m]	M1 M1A1 M1A1 [5]
(c)	$\frac{\mathrm{d}v}{\mathrm{d}t} = 2t - 10$ when $t = 9$, acceleration = 8 [m/s ^{2}]	M1 A1 [2]
(d)	(i) $v = (t-5)^2 + 3$ Irrespective of the value of t $v \geqslant 3$ so the particle never comes to rest. **ALT** $b^2 - 4ac < 0 \Rightarrow (-10)^2 - 4\times1\times28 = -12$ No real solutions so the particle never comes to rest. (ii) Least value of v is 3 [m/s]	M1 A1 [M1 A1] B1 [3]
		Total 11 marks

Question	Notes	Marks
7(a)	$v = t^2 - 10t + 28$	
	$v = 1^2 - 10\times1 + 28 = 19$ [m/s]	B1 [1]
(b)	For an attempt to integrate the given expression for v [See general guidance for the definition of an attempt] $s = \int\left(t^2 - 10t + 28\right)\,\mathrm{d}t = \frac{t^3}{3} - \frac{10t^2}{2} + 28t\,(+c)$	M1
	For finding the value of c. They cannot score this mark without $+c$ $24 = \frac{3^3}{3} - \frac{10\times3^2}{2} + 28\times3 + c \Rightarrow c = \ldots$	M1
	For the correct expression for s. This need not be explicitly stated. $s = \frac{t^3}{3} - \frac{10t^2}{2} + 28t - 24$	A1

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Answer 8 of 8 · Question 7 part b · 5 marks

2023 JAN · Paper 01 (regional R) · extract 2 of 3 · find part b in this mark scheme extract

	Award for the correct value of c seen -24 [m]	
	For using their integrated expression for s to find a value of the displacement when $t = 5$ $s = \frac{5^3}{3} - \frac{10\times5^2}{2} + 28\times5 - 24 = ...$	M1
	For the correct value of $s = \frac{98}{3}$ [m] Accept $s = 32.7$ [or better]	A1 [5]
(c)	For an attempt to differentiate the given v and substituting $t = 9$ into their differentiated expression. $\frac{dv}{dt} = 2t - 10 \Rightarrow \frac{dv}{dt} = 2\times9 - 10 = ...$	M1
	For acceleration = 8 [m/s ^{2}]	A1 [2]
(d)(i)	**Method A** Completes the square to give $v = (t-5)^2 + 3$	M1
	Concludes that at the minimum velocity [3 m/s] $t = 5$ so P never comes to rest	A1
	Method B Finds the value of the discriminant $b^2 - 4ac < 0 \Rightarrow (-10)^2 - 4\times1\times28 = -12$	M1
	Concludes that as there are no real solutions, **so *P* does not come to rest.**	A1
	Method C Solves the 3TQ to give the following two [non-real] values of t: $5+\sqrt{3}\mathrm{i}$ and $5-\sqrt{3}\mathrm{i}$ or $\left(t-\left[5+\sqrt{3}\mathrm{i}\right]\right)\left(t-\left[5+\sqrt{3}\mathrm{i}\right]\right)=0$	M1
	Concludes that as there are no real solutions, **so *P* does not come to rest.**	A1
	Method D Uses their result from (c) $\frac{dv}{dt} = 2t - 10 = 0 \Rightarrow t = 5$	M1
	Concludes that at the minimum velocity [3 m/s] $t = 5$ so P never comes to rest	A1
(ii)	For the correct value of $v = 3$ [m/s}	B1 [3]
		Total 11 marks

Type MS · Velocity and displacement by integrating

Answer 8 of 8 · Question 7 part b · 5 marks

2023 JAN · Paper 01 (regional R) · extract 3 of 3 · find part b in this mark scheme extract

Question	Scheme	Marks